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18.01SC

Single Variable Calculus 18.01SC — Single Variable Calculus May 2026

EXTREMELY HARD – MIT LEVEL**Time Limit: 3 hours****Total Marks: 100**

1. (3 points) Prove the Brouwer fixed-point theorem using differential topology: any continuous map $f : \overline{B}^n \rightarrow \overline{B}^n$ has a fixed point. Use the fact that there is no smooth retraction of the ball onto its boundary.
2. (3 points) Prove the open mapping theorem: if $T : X \rightarrow Y$ is a bounded linear operator between Banach spaces and T is surjective, then T maps open sets to open sets. Use it to prove the inverse mapping theorem.
3. (5 points) Prove the Nash embedding theorem for compact Riemannian manifolds: any compact C^∞ Riemannian manifold can be isometrically embedded into $\mathbb{R}^N$ for sufficiently large N .
4. (2 points) Prove the Schauder fixed-point theorem using Schauder estimates and the Arzela-Ascoli theorem: a compact operator $T : X \rightarrow X$ on a Banach space has a fixed point if it maps a closed convex set into itself.
5. (5 points) Prove the Poincaré inequality: for $u \in W_0^{1,p}(\Omega)$ with $\Omega \subset \mathbb{R}^n$ bounded, $\|u\|_{L^p} \leq C \|\nabla u\|_{L^p}$. Show that the constant C depends on the diameter of Ω and give the optimal constant for $p = 2$ on an interval.
6. (3 points) Prove the Gagliardo-Nirenberg-Sobolev inequality: for $1 \leq p < n$, there exists $C = C(n, p) > 0$ such that $\|u\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|\nabla u\|_{L^p(\mathbb{R}^n)}$ for all $u \in C_c^1(\mathbb{R}^n)$, where $p^* = np/(n - p)$.
7. (4 points) Let M be a compact oriented n -manifold with boundary ∂M . State and prove Stokes' theorem for differential forms. Then apply it to prove the divergence theorem: $\int_M \operatorname{div} F \, dV = \int_{\partial M} F \cdot n \, dS$.
8. (5 points) Prove the Lax-Milgram theorem and use it to show existence and uniqueness of weak solutions to the Dirichlet problem $-\Delta u = f$ in Ω , $u|_{\partial\Omega} = 0$ for $f \in L^2(\Omega)$.
9. (3 points) Prove that $\mathbb{R}P^3$ is diffeomorphic to $SO(3)$ by constructing explicit maps and verifying they are smooth inverses. Use this to compute $\pi_1(SO(3))$.

10. (5 points) Prove that the Laplace operator Δ on a compact Riemannian manifold has discrete spectrum $\lambda_1 \leq \lambda_2 \leq \dots \rightarrow \infty$ and each eigenvalue has finite multiplicity. Show $\lambda_1 = \inf \frac{\|\nabla u\|^2}{\|u\|^2}$ over nonzero $u \in H^1(M)$ with $\int u = 0$.
11. (4 points) Prove the Hopf-Rinow theorem: for a Riemannian manifold (M, g) , metric completeness is equivalent to geodesic completeness (every geodesic extends for all time).
12. (3 points) Prove using the epsilon-delta definition that the function $f(x) = x^2 \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$ is differentiable at $x = 0$ but $f'(x)$ is not continuous at $x = 0$. Carefully justify all steps in your epsilon-delta arguments.
13. (5 points) Prove the Atiyah-Singer index theorem for a simple case: compute the index of the Dirac operator on S^2 and verify it equals the integral of the $\hat{A}$ -genus.
14. (5 points) Let $f_n : [0, 1] \rightarrow \mathbb{R}$ be a sequence of functions of bounded variation with $\text{Var}(f_n) \leq C$ uniformly and $f_n \rightarrow f$ pointwise. Prove that f is of bounded variation and $\text{Var}(f) \leq C$. Give a counterexample showing that uniform bounded variation is necessary.
15. (4 points) Construct an exhaustion function (proper smooth function bounded below) on any smooth manifold and use it to prove that every smooth manifold admits a complete Riemannian metric.
16. (2 points) Let (X, d) be a complete metric space and $T : X \rightarrow X$ a contraction mapping with contraction factor $k < 1$. Prove the Banach fixed-point theorem: T has a unique fixed point. Additionally, derive the error estimate $\|x_n - x^*\| \leq \frac{k^n}{1-k} \|x_1 - x_0\|$ where $x_{n+1} = T(x_n)$.
17. (5 points) Prove the maximum principle for elliptic PDEs: if $Lu = -\Delta u + c(x)u \geq 0$ in Ω with $c \geq 0$, then $\max_{\overline{\Omega}} u \leq \max_{\partial\Omega} u^+$. Show how this implies uniqueness for the Dirichlet problem.
18. (5 points) Construct a sequence of continuous functions $f_n : [0, 1] \rightarrow \mathbb{R}$ that converges pointwise to $f(x) = 0$ but does NOT converge uniformly. Prove your construction works and explain why uniform convergence fails.
19. (5 points) Analyze the stability of the Runge-Kutta method of order 4 when applied to the stiff ODE $y' = \lambda y$ with $\lambda \in \mathbb{C}$, $\text{Re}(\lambda) < 0$. Determine the region of absolute stability and discuss how the step size h must be chosen for stiff problems.
20. (4 points) State and prove the spectral theorem for compact self-adjoint operators on a Hilbert space. Show that such an operator has a countable set of eigenvalues that accumulate only at zero, and the eigenvectors form an orthonormal basis.

21. (3 points) Classify the second-order linear PDE $u_{xx} + 2u_{xy} + u_{yy} + u_x + u_y = 0$. Determine its type (elliptic, parabolic, hyperbolic) and find its canonical form. Solve the resulting canonical equation.
22. (6 points) Prove the Hille-Yosida theorem: a linear operator A generates a strongly continuous contraction semigroup on a Banach space X iff A is closed, densely defined, and all $\lambda > 0$ are in the resolvent set with $\|R(\lambda, A)\| \leq 1/\lambda$.
23. (2 points) Prove the Rellich-Kondrachov compactness theorem: $W^{1,p}(\Omega) \hookrightarrow L^q(\Omega)$ compactly for $1 \leq q < p^*$ when Ω is bounded and $\partial\Omega$ is C^1 .
24. (5 points) Prove that there is no continuous linear functional $L : \ell^\infty \rightarrow \mathbb{R}$ that extends the limit functional on c (convergent sequences) and satisfies $\|L\| = 1$. (Hint: use the Hahn-Banach theorem and consider the Banach limit.)
25. (4 points) Compute the Fourier transform of the distribution $p.v.(1/x)$ and prove that $\widehat{p.v.(1/x)}(\xi) = -i\pi \operatorname{sgn}(\xi)$. Use this to solve the Hilbert transform equation $H(u) = f$.

Solutions

Solution 1:

Suppose f has no fixed point. Define $r(x)$ = intersection of ray from $f(x)$ through x with ∂B^n . This gives a smooth retraction (after smooth approximation). By Sard's theorem, regular values exist. For a regular value y , $r^{-1}(y)$ is a 1-manifold with boundary, one endpoint at y . The other endpoint would be in B^n , impossible since r maps interior to boundary. Contradiction.

Solution 2:

Use Baire category: $Y = \cup_n nT(B_X(0, 1))$ so some $nT(B_X(0, 1))$ has nonempty interior. Show $T(B_X(0, 1))$ contains a neighborhood of 0. Then by scaling, T maps open balls to open sets. The inverse mapping theorem follows: if T is bijective bounded linear, then T^{-1} is bounded.

Solution 3:

Sketch: Use the implicit function theorem to construct a short immersion $f_0 : M \rightarrow \mathbb{R}^N$. The metric g on M can be expressed as $f_0^* g_{\mathbb{R}^N} + h$. Use the Nash-Moser iteration to solve the system of PDEs $\partial_i f \cdot \partial_j f = g_{ij}$. The iteration converges by tame estimates on the linearized operator $L_f(h) = df \cdot \nabla h$.

Solution 4:

Let $C \subset X$ be closed convex, $T(C) \subset C$, T compact. Approximate T by finite-rank operators T_n using Schauder basis. By Brouwer, each T_n has a fixed point x_n . By compactness of T , $\{T(x_n)\}$ has convergent subsequence $T(x_{n_k}) \rightarrow y$. Since $x_{n_k} = T_{n_k}(x_{n_k})$ and $T_{n_k} \rightarrow T$ uniformly, y is a fixed point of T .

Solution 5:

For $u \in C_c^\infty(\Omega)$, write $u(x) = \int_{-\infty}^{x_1} \partial_1 u(t, x'') dt$. By Hölder, $|u(x)|^p \leq (\text{diam } \Omega)^{p-1} \int |\partial_1 u|^p dt$. Integrate over Ω to get $\|u\|_p^p \leq (\text{diam } \Omega)^p \|\nabla u\|_p^p$. On $(0, \pi)$, the optimal constant for $p = 2$ is $C = 1/\pi$, attained by $u(x) = \sin(x)$.

Solution 6:

For $p = 1$, use $|u(x)| \leq \int_{-\infty}^{x_i} |\partial_i u| dx_i$, multiply over $i = 1, \dots, n$, integrate, apply Hölder repeatedly to get $\|u\|_{n/(n-1)} \leq \prod (\int |\partial_i u|)^{1/n} \leq \frac{1}{n} \sum \int |\partial_i u|$. For general p , apply the $p = 1$ case to $|u|^q$ with suitable q and use Hölder.

Solution 7:

Stokes: $\int_M d\omega = \int_{\partial M} \omega$ for an $(n-1)$ -form ω . Proof: Use partition of unity to reduce to $\mathbb{R}^n$ case, where it follows from the fundamental theorem of calculus and Fubini. For divergence theorem, let $\omega = \sum_i (-1)^{i-1} F_i dx^1 \wedge \dots \wedge \widehat{dx^i} \wedge \dots \wedge dx^n$. Then $d\omega = (\sum_i \partial F_i / \partial x^i) dx^1 \wedge \dots \wedge dx^n = \text{div } F dV$. The boundary term gives $\int_{\partial M} F \cdot n dS$ by the Hodge star duality.

Solution 8:

Lax-Milgram: Let $a : H \times H \rightarrow \mathbb{R}$ be coercive bilinear continuous on Hilbert H , $f \in H^*$. Then $\exists! u \in H$: $a(u, v) = f(v)$ for all $v \in H$. For the Dirichlet problem, $H = H_0^1(\Omega)$, $a(u, v) = \int_\Omega \nabla u \cdot \nabla v$, $f(v) = \int_\Omega f v$. Poincaré inequality gives coercivity,

trace theorem gives boundary values.

Solution 9:

Map $\mathbb{R}P^3 \rightarrow SO(3)$: represent a point in $\mathbb{R}P^3$ as a unit quaternion q modulo sign. The conjugation map $v \mapsto qvq^{-1}$ on pure quaternions gives a rotation in $SO(3)$. This is a smooth double cover $S^3 \rightarrow SO(3)$ with kernel $\{\pm 1\}$, inducing $\mathbb{R}P^3 \cong SO(3)$. Thus $\pi_1(SO(3)) \cong \mathbb{Z}/2\mathbb{Z}$.

Solution 10:

The resolvent $(-\Delta + I)^{-1} : L^2 \rightarrow H^2$ is compact by Rellich-Kondrachov. The spectral theorem for compact self-adjoint operators gives discrete eigenvalues $\mu_n \rightarrow 0$, so $\lambda_n = 1/\mu_n - 1 \rightarrow \infty$. Finite multiplicity follows from finite-dimensional eigenspaces of compact operators. The variational characterization follows from the Rayleigh quotient. The condition $\int u = 0$ removes the trivial $\lambda_0 = 0$.

Solution 11:

($\Rightarrow$): Assume metric completeness. For a geodesic γ with maximal interval (a, b) , if $b < \infty$, take $t_n \rightarrow b$. Then $\{\gamma(t_n)\}$ is Cauchy (distance $\leq C|t_n - t_m|$) so converges. Extend γ using the exponential map at the limit point. ($\Leftarrow$): Assume geodesic completeness. Take a Cauchy sequence p_n . Connect each p_n to p_0 by a minimizing geodesic. Bound on distances and completeness of initial data gives convergent subsequence.

Solution 12:

We prove $f'(0) = 0$ by showing $|(f(h) - f(0))/h - 0| = |h \sin(1/h)| \leq |h|$. For any $\varepsilon > 0$, choose $\delta = \varepsilon$. Then $|h| < \delta$ implies $|h \sin(1/h)| < \varepsilon$, so $f'(0) = 0$. For $x \neq 0$, $f'(x) = 2x \sin(1/x) - \cos(1/x)$. This oscillates wildly near 0 since $\cos(1/x)$ has no limit. Hence f' is not continuous at 0.

Solution 13:

The Dirac operator D on S^2 acts on spinors. $\text{ind}(D) = \dim \ker D - \dim \ker D^*$. By Lichnerowicz formula, $D^2 = \nabla^* \nabla + \frac{1}{4}R$. On S^2 with $R = 2$, $\ker D$ consists of the constant spinor. So $\text{ind}(D) = 1$. The $\hat{A}$ -genus: $\int_{S^2} \hat{A}(TS^2) = \int_{S^2} 1 - \frac{p_1}{24} + \cdots = 1$ since $p_1(TS^2) = 2\chi = 4$ but the integral gives 0 at this order. Indeed $\int_{S^2} \frac{R}{4\pi} = \chi/2 = 1$.

Solution 14:

For any partition $0 = x_0 < \cdots < x_k = 1$, $|f(x_i) - f(x_{i-1})| = \lim_n |f_n(x_i) - f_n(x_{i-1})| \leq \liminf_n \text{Var}(f_n) \leq C$. So $\text{Var}(f) \leq C$. Counterexample without UB: $f_n(x) = nx$ on $[0, 1/n]$, $f_n(x) = 1$ elsewhere. $\text{Var}(f_n) = 2n$, $f_n \rightarrow 1$ pointwise, but limit has variation 0, so inequality fails direction.

Solution 15:

Let $\{U_i\}$ be a countable cover by coordinate balls, $\{\rho_i\}$ a partition of unity. Define $f = \sum_i \rho_i$. This is proper: $f^{-1}([-\infty, c])$ is compact (only finitely many $i \leq c$ contribute). For a Riemannian metric, start with any metric g_0 , let $g = g_0 + df \otimes df$ on $f^{-1}((-\infty, \infty))$. The gradient of f has unit length, so any curve must escape to infinity in infinite length.

Solution 16:

Pick any $x_0 \in X$, define $x_{n+1} = T(x_n)$. Then $d(x_{n+1}, x_n) \leq k^n d(x_1, x_0)$. By triangle

inequality, $d(x_{n+m}, x_n) \leq \sum_{i=0}^{m-1} k^{n+i} d(x_1, x_0) \leq \frac{k^n}{1-k} d(x_1, x_0)$. So $\{x_n\}$ is Cauchy, converges to x^* . By continuity of T , $T(x^*) = x^*$. Uniqueness: if $Tx = x$, $Ty = y$, then $d(x, y) = d(Tx, Ty) \leq kd(x, y)$, so $d(x, y) = 0$. The error estimate follows from taking $m \rightarrow \infty$.

Solution 17:

Suppose u attains an interior maximum at x_0 with $u(x_0) = M > 0$ and $M > \max_{\partial\Omega} u^+$. At x_0 , $\Delta u \leq 0$ (Hessian negative semidefinite), so $Lu = -\Delta u + cu \geq 0 + cM \geq 0$. Strictly, if $c > 0$ or $M > 0$, this gives $Lu > 0$ at the maximum, contradiction. For uniqueness of $\Delta u = f$ with $u|_{\partial\Omega} = g$, difference of two solutions satisfies $\Delta w = 0$, $w|_{\partial\Omega} = 0$, so $w \equiv 0$.

Solution 18:

Let $f_n(x) = \max(0, 1 - n|x - 1/n|)$ (triangular spikes at $1/n$). For any x , for $n > 1/x$, $f_n(x) = 0$, so $f_n \rightarrow 0$ pointwise. But $\|f_n\|_\infty = 1$ for all n , so $\sup_{x \in [0,1]} |f_n(x) - 0| = 1 \not\rightarrow 0$, hence convergence is not uniform. This shows pointwise convergence does not imply uniform convergence.

Solution 19:

The RK4 method applied to $y' = \lambda y$ gives $y_{n+1} = R(z)y_n$ where $z = h\lambda$ and $R(z) = 1 + z + z^2/2 + z^3/6 + z^4/24$. The stability region is $\{z \in \mathbb{C} : |R(z)| \leq 1\}$. For stiff problems where $\text{Re}(\lambda)$ is large negative, $|R(z)|$ grows for large $|z|$ (RK4 is not A-stable), requiring h small enough that $h\lambda$ stays in the stability region.

Solution 20:

Let $T : H \rightarrow H$ be compact self-adjoint. Show $\|T\| = \sup_{|x|=1} |\langle Tx, x \rangle|$ (achieved). Let $\lambda_1 = \pm\|T\|$ be the extremal eigenvalue with eigenvector e_1 . Inductively apply the same argument to T restricted to $\{e_1, \dots, e_{n-1}\}^\perp$. Compactness implies $\lambda_n \rightarrow 0$. The eigenvectors form an ONB by the spectral decomposition $Tx = \sum \lambda_n \langle x, e_n \rangle e_n$.

Solution 21:

The characteristic equation: $a_{11}dy^2 - 2a_{12}dxdy + a_{22}dx^2 = 0$ gives $1 \cdot dy^2 - 2 \cdot 1 \cdot dxdy + 1 \cdot dx^2 = 0$, so $(dy - dx)^2 = 0$. Double root: parabolic type. Characteristic curves: $y - x = c$. Canonical coordinates: $\xi = y - x$, $\eta = x$. Then $u_{\eta\eta} + u_\eta = 0$. Solve: $u(\xi, \eta) = A(\xi)e^{-\eta} + B(\xi)$, i.e. $u(x, y) = A(y - x)e^{-x} + B(y - x)$.

Solution 22:

($\Rightarrow$): If A generates $T(t)$, show $R(\lambda, A) = \int_0^\infty e^{-\lambda t} T(t) dt$, so $\|R(\lambda, A)\| \leq \int_0^\infty e^{-\lambda t} \|T(t)\| dt \leq 1/\lambda$. ($\Leftarrow$): Define $A_n = nAR(n, A)$ (Yosida approximants) which are bounded generators of $T_n(t) = e^{tA_n}$. Show $T_n(t)$ converges strongly to a semigroup $T(t)$ and its generator is A .

Solution 23:

Take a bounded sequence $\{u_n\}$ in $W^{1,p}$. By the Sobolev-Gagliardo-Nirenberg inequality, it's bounded in L^{p^*} . Use the Frechet-Kolmogorov criterion for compactness in L^q : show $\|u_n(\cdot + h) - u_n\|_{L^q} \rightarrow 0$ uniformly as $h \rightarrow 0$ using difference quotients and the $W^{1,p}$ bound. Extract a subsequence convergent in L^q by the Arzela-Ascoli type argument.

Solution 24:

The Hahn-Banach theorem guarantees the existence of such extensions (Banach limits), but they are not unique and not continuous in the sense of being representable by a sequence. The key issue: ℓ^∞/c_0 is non-separable and the quotient map is not injective. Banach limits exist but require the axiom of choice. They satisfy $\liminf x_n \leq L(x) \leq \limsup x_n$ and are shift-invariant.

Solution 25:

$\langle \widehat{p.v.(1/x)}, \varphi \rangle = \langle p.v.(1/x), \widehat{\varphi} \rangle$. Write $p.v.(1/x) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{x} \chi_{|x| > \varepsilon}$. Its Fourier transform: $\int_{|x| > \varepsilon} \frac{e^{-2\pi i x \xi}}{x} dx = -i\pi \operatorname{sgn}(\xi)$ as $\varepsilon \rightarrow 0$. The Hilbert transform $H(u) = \frac{1}{\pi} p.v. \int \frac{u(t)}{x-t} dt$ has symbol $\widehat{H(u)}(\xi) = -i \operatorname{sgn}(\xi) \widehat{u}(\xi)$, so $H^{-1} = -H$.